Gitesh Kumar

AI & Operations Engineer | Industrial Systems & Process Improvement

AI Engineer with 4+ years of hands-on industrial experience at a large-scale energy facility and a Master's from FAU Erlangen-Nürnberg. During my time at HPCL I initiated the digitisation of plant workflows from scratch which delivered a significant improvement in production efficiency. I am now looking to explore how modern AI can drive that kind of operational impact at a much larger scale and much faster rate.

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EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg — From October 2022 to March 2026
Masters in Elektromobilität-ACES
Specialization: AI and Sustainable Production Technology

PEC University of Technology Chandigarh, India — From July 2013 to May 2017
Bachelors of Engineering
Mechanical Engineering

WORK EXPERIENCE

Sr. Operations Officer — From July 2017 to August 2021
Hindustan Petroleum Corporation Limited India

- Managed day-to-day operations of a high-capacity LPG bottling plant, ensuring reliability of critical systems and acting as the main contact point with Honeywell to keep automation projects aligned with operational needs.
- Replaced manual reporting with structured digital workflows by integrating SCADA sensor data and ERP systems to give all teams accurate visibility into plant performance for the first time.
- Partnered with maintenance and safety teams to build datasets from operational sensor data that supported production monitoring and helped identify recurring inefficiencies before they escalated.
- Developed performance reporting tools that translated raw machine data into actionable insights, contributing to a 10% improvement in production efficiency through faster and better informed operational decisions.

RECENT ACADEMIC PROJECTS

Generative AI Agent for Industrial Engineering

- Built a full-stack industrial data pipeline that transforms raw factory sensor data into financial risk insights using a Medallion architecture to move from raw sensor data to business decisions in a structured and auditable way.
- Developed an AI agent that acts as a virtual plant manager. Engineers can ask it natural language questions like "which machine should I repair first" and it reasons through the data, then writes its own SQL queries and returns a factually grounded answer.
- Designed the entire system to run locally without sending sensitive operational data to external cloud providers because in industrial environments data sovereignty is not optional.

Master's Thesis: A Retrieval-Augmented Framework for Fine-tuned LLMs for Vulnerability Analysis

- Developed and evaluated a retrieval-augmented LLM pipeline for cybersecurity analysis using LLaMA-based models and domain-specific datasets.
- Analyzed strengths, limitations and failure modes of domain-adapted LLMs in vulnerability and cybersecurity reasoning.
- The focus was on de-risking the technology by conducting systematic stress tests, proving that we can make advanced AI reliable enough for a global enterprise.
- Designed controlled experiments comparing baseline and RAG-enhanced models using NLP evaluation metrics and statistical testing.

SKILLS

Python, SQL, SCADA systems, ERP integration, LLMs, RAG, Process Improvement, Data-Driven Decision Making, System Thinking, Problem Structuring, Cross-functional Collaboration

LANGUAGES

English
IELTS: 8/9

German
B1-(actively improving)

SOCIAL NETWORKS

@gitesh-kumar
Gitesh,Kumar